

Question	Answer	Marks
4(a)(i)	(a wave that) transfers / propagates energy	B1
4(a)(ii)	distance = $n\lambda$	B1
	time $t = n / f$	B1
4(a)(iii)	(v = distance / time) (v) = $n\lambda / (n / f)$ so (v) = $f\lambda$	B1
4(b)(i)	$f = 3 \times 10^8 / 4 \times 10^{-2}$ $= 7.5 \times 10^9$ (Hz) $= 7.5 \times 10^9 / 10^9$ (GHz)	C1
	= 7.5 GHz	A1
4(b)(ii)	(difference in path lengths, XQ – YQ) = $\left[72^2 + 30^2\right]^{\frac{1}{2}} - 72 = 6$ (cm)	A1
	Or calculate angle Q using tan and then cos to obtain XQ of 78 $\tan \theta = 30 / 72$ so $\theta = 22.6$ $XQ = 72 / \cos 22.6 = 78$ (cm) path difference $78 - 72 = 6$ (cm)	(A1)
	Or calculate angle X using tan and then cos to obtain XQ of 78 $\tan \theta = 72 / 30$ so $\theta = 67.4$ $XQ = 72 / \sin 67.4 = 78$ (cm) path difference $78 - 72 = 6$ (cm)	(A1)
4(b)(iii)	path difference (= 6 cm / 4 cm) $\lambda = 1.5\lambda$	M1
	(so) phase difference (at Q) = 540° or 180°	M1
	(so) intensity is minimum	A1

Question	Answer	Marks
4(b)(iv)	amplitude changes from maximum (at P) to minimum (at Q)	C1
	amplitude changes from maximum (at P) to minimum to maximum to minimum (at Q)	A1

Question	Answer	Marks
5(c)(i)	temperature decreases, resistance decreases	B1